

Ridge Regression

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Introduction

Ridge regression adds an L2 penalty $\lambda \sum_j \beta_j^2$ to the residual sum of squares. It shrinks coefficients toward zero — but never exactly to zero — reducing variance at the cost of a small bias and producing stable fits even when predictors are highly correlated. The classical motivation was multicollinearity: when $X^\top X$ is near-singular, OLS estimates are extremely variable, and the ridge penalty regularises the inverse and yields a well-defined estimator. Modern uses extend to high-dimensional regression where $p > n$ makes OLS undefined.

Prerequisites

A working understanding of linear regression, the bias-variance trade-off, multicollinearity, and the role of cross-validation in tuning regularisation parameters.

Theory

The ridge estimator minimises

$$\|y - X\beta\|^2 + \lambda\|\beta\|_2^2,$$

with closed-form solution $\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$. Larger λ shrinks coefficients more aggressively; $\lambda = 0$ recovers OLS. The coefficient paths — coefficient values as λ varies on a log scale — show how each predictor contributes; coefficients shrink continuously rather than dropping out abruptly as in lasso.

Cross-validation (typically $k = 10$ folds) selects λ . Two common rules: $\lambda_{\min}$ minimises CV error, λ_{1se} takes the largest λ within one standard error of the minimum (more parsimonious shrinkage).

Assumptions

Standard linear-model assumptions on residuals; predictors should be standardised (`glmnet` does this internally and back-transforms). For statistical inference on shrunk coefficients, the bias from regularisation must be considered; bootstrap or debiased lasso/ridge methods provide adjusted intervals.

R Implementation

```
library(glmnet)

X <- as.matrix(mtcars[, -1])
y <- mtcars$mpg

set.seed(2026)
cvfit <- cv.glmnet(X, y, alpha = 0)  # alpha = 0 -> ridge

plot(cvfit)
```

```
coef(cvfit, s = "lambda.min")
coef(cvfit, s = "lambda.1se")
```

Output & Results

`cv.glmnet()` with `alpha = 0` fits ridge regression with cross-validated λ . The plot shows mean CV error against $\log \lambda$ with the two selection lines. `coef()` returns shrunk coefficients at the chosen λ ; all are non-zero, in contrast to lasso.

Interpretation

A reporting sentence: “Ridge regression with CV-optimal $\lambda_{1se} = 0.5$ produced shrunk coefficient estimates that remained stable despite the strong correlations between displacement, weight, and horsepower; the cross-validated mean squared error was 7.2 mpg² versus 9.8 mpg² for OLS, reflecting the variance reduction.” Always report the chosen λ and the cross-validated error.

Practical Tips

- Always standardise predictors before ridge; `glmnet` does this internally with `standardize = TRUE` (default).
- Use λ_{1se} for more shrinkage than λ_{min} ; the parsimonious choice typically generalises better.
- For variable selection rather than pure shrinkage, switch to lasso (`alpha = 1`) or elastic net (`0 < alpha < 1`); ridge keeps every predictor.
- For non-Gaussian outcomes, set `family = "binomial", "poisson",` etc.; the ridge penalty extends naturally.
- Set the random seed before `cv.glmnet()` to ensure reproducible λ selection.
- Compare ridge against lasso and elastic net on the same cross-validation folds when the right level of sparsity is unclear; modern penalised regression workflows benchmark all three.

R Packages Used

`glmnet` for canonical ridge, lasso, and elastic-net implementations with cross-validated tuning; `MASS::lm.ridge()` for an alternative ridge implementation; `hdi` for high-dimensional inference including debiased ridge; `caret` and `tidymodels` for unified cross-validated comparison across penalised methods.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (`emmeans`)
- Two-way / factorial ANOVA with interaction

- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI